

§5. 一元函数微分学的应用 — 几何应用

一. 极值与单调性. [例5.7~5.13]

1. 费马定理: 极值则导数为0.

2. 左 $f'(x) < 0$ 右 $f'(x) > 0$. $x=x_0$ 极小...

3) $\Rightarrow$ 单调导 $f'(x_0)=0$ $f''(x_0) > 0$, 极小.

4) n 阶同导 $\sim n-1$ 阶同导 $f^{(n)}(x_0) > 0$, 极小.
且 n 是偶数.

二. 拐点与凹凸性. [例5.16~5.14]

凹:   

三. 渐近线. [例5.17~5.18]

水平 $x \rightarrow x_0 \neq \infty$
水平 $x \rightarrow \pm\infty = C$
斜: $\frac{f(x)}{x} \rightarrow k$ $f(x) - kx \rightarrow b$.

四. 最值略

五. 曲率: $k = \frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}}$ 曲率半径 $\frac{1}{k} = R$.

六. 相关变化率:

$$\frac{dA}{dB} = \frac{dA}{dC} \cdot \frac{dC}{dB}. \quad [\text{例5.24}]$$

例 5.3

曲线极坐标方程 $r = 1 - \cos\theta$

求该曲线上对应于 $\theta = \frac{\pi}{6}$ 处的切线方程

[分析] $r = 1 - \cos\theta$

$$\begin{cases} x = r(\theta) \cos\theta = (1 - \cos\theta) \cos\theta \\ y = r(\theta) \sin\theta = (1 - \cos\theta) \sin\theta \end{cases}$$

$$k = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = -\frac{\cos\theta - \cos 2\theta}{\sin\theta - \sin 2\theta}$$

[注: $r = 1 + \cos\theta$, $k = -\frac{\cos\theta + \cos 2\theta}{\sin\theta + \sin 2\theta}$]

$k|_{\theta=\frac{\pi}{6}} = 1 \Rightarrow y - y_0 = k(x - x_0)$

$x_0 = \frac{\pi}{6} = \frac{\sqrt{3}}{2} - \frac{1}{4}$ 则 $y - (\frac{1}{2} - \frac{\sqrt{3}}{4}) = x - (\frac{\sqrt{3}}{2} - \frac{1}{4})$

$y_0 = \frac{\pi}{6} = \frac{1}{2} - \frac{\sqrt{3}}{4}$ 则 $y - (\frac{1}{2} - \frac{\sqrt{3}}{4}) = x - (\frac{\sqrt{3}}{2} - \frac{1}{4})$

例 5.1

$f(x)$ 是周期为 π 的连续函数它在 $x=0$ 某邻域内:

$$f(1+\sin x) - 3f(1-\sin x) = 8x + \alpha(x)$$

其中 $\alpha(x)$ 是当 $x \rightarrow 0$ 时比 x 高阶的无穷小, 且 $f(x)$ 在 $x=1$ 处可导, 求 $y=f(x)$ 在 $(1, f(1))$ 处切线方程

$$f(1) = f(1) \quad f'(1) = f'(1)$$

$$\lim_{x \rightarrow 0} f(1) - 3f(1) = 0 + \alpha(x) \Rightarrow f(1) = 0$$

$$\lim_{x \rightarrow 0} \frac{f(1+\sin x) - 3f(1-\sin x)}{\sin x} = f'(1) + 8x$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{f(1+\sin x) - 3f(1-\sin x)}{\sin x} = \lim_{x \rightarrow 0} \frac{8x + \alpha(x)}{\sin x} = 8$$

$$f'(1) = \lim_{x \rightarrow 0} \frac{f(1+\sin x) - f(1)}{\sin x}$$

$$= 8 + \lim_{x \rightarrow 0} \frac{3f(1-\sin x) - f(1)}{\sin x} = 8 + \lim_{x \rightarrow 0} \frac{3f(1-\sin x) - 3f(1)}{\sin x}$$

$$\Rightarrow 4f'(1) = 8 \Rightarrow f'(1) = 2$$

切线方程 $x - 1 = 2(y - 0) \quad y - 0 = 2(x - 1)$

例 5.2

$x = \int_0^{1-t} e^{-u^2} du$ 在 $(0, 1)$ 上切线方程

$y = t^2 \ln(2-t^2)$

$$\frac{dy}{dx} = \frac{2t \ln(2-t^2) + \frac{t^2 \cdot (-2t)}{2-t^2}}{e^{-(1-t)^2} \cdot (-1)}$$

$y=0 \Rightarrow t=0$ 或 $t=\pm 1$

$\int_0^1 e^{-u^2} du \neq 0 \quad \int_0^2 e^{-u^2} du \neq 0 \Rightarrow t=1$

$k = \frac{0 + (-2)}{-1} = 2 \quad y = 2x$

例 5.4 设 $y=f(x)$ 由方程 $e^{2x+y} - \cos xy = e - 1$ 所确定

求 $y=f(x)$ 在 $(0, 1)$ 处切线方程

$$e^{2x+y}(2+y') + \sin xy \cdot (y + xy') = 0$$

$x=0, y=1 \quad y'=-2 \quad y-1 = -2(x-0)$

例 5.5

曲线 $(2-x^{n^2})y=1$ 在 $(1,1)$ 处切线 与 x 轴交点为

$$(x_n, 0), n=2,3,\dots, \lim_{n \rightarrow \infty} x_n = ?$$

截距可正可负

[分析] 函数族 (曲线族).

$$y = \frac{1}{2-x^{n^2}} \quad y' = -\frac{1}{(2-x^{n^2})^2} \cdot (-n^2) x^{n^2-1}$$

$$y'|_{x=1} = n^2 = k(n).$$

$$\text{切线族为 } y-1 = k(n) \cdot (x-1) = n^2(x-1)$$

$$\lim_{n \rightarrow \infty} (x_n)^{\frac{n^2}{2}} = \lim_{n \rightarrow \infty} e^{\frac{n^2}{2}(x_n-1)} = \lim_{n \rightarrow \infty} e^{\frac{n^2}{2}(-\frac{1}{n^2})} = e^{-\frac{1}{2}}$$

$$(u^v)_{u \rightarrow \infty} = e^{v(u-1)}$$

例 5.6

证明 曲线弧 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} (x>0, y>0, a>0)$

的切线在两坐标轴上的截距平方之和为常数

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}} \cdot y' = 0 \quad x^{-\frac{1}{3}} + y^{-\frac{1}{3}} \cdot y' = 0$$

$$\Rightarrow y' = -\frac{x_0^{-\frac{1}{3}}}{y_0^{-\frac{1}{3}}}, \quad x_0^{\frac{2}{3}} + y_0^{\frac{2}{3}} = a^{\frac{2}{3}}, \quad y - y_0 = -\frac{x_0^{-\frac{1}{3}}}{y_0^{-\frac{1}{3}}}(x - x_0)$$

$$\begin{cases} \text{令 } x=0 & y_n = \frac{x_0^{\frac{2}{3}}}{y_0^{-\frac{1}{3}}} + y_0 = y_0(y_0^{\frac{2}{3}} + x_0^{\frac{2}{3}}) \\ \text{令 } y=0 & x_n = \frac{y_0^{\frac{2}{3}}}{x_0^{-\frac{1}{3}}} + x_0 = x_0(x_0^{\frac{2}{3}} + y_0^{\frac{2}{3}}) \end{cases}$$

$$x_n^2 + y_n^2 = x_0^2 + y_0^2 + 2x_0^{\frac{2}{3}}y_0 + 2y_0^{\frac{2}{3}}x_0 + y_0x_0^{\frac{4}{3}} + x_0y_0^{\frac{4}{3}}$$

$$\begin{aligned} &= y_0^2 (y_0^{\frac{4}{3}} + x_0^{\frac{4}{3}} + 2y_0^{\frac{2}{3}}x_0^{\frac{2}{3}}) + x_0^2 (x_0^{\frac{4}{3}} + y_0^{\frac{4}{3}} + 2x_0^{\frac{2}{3}}y_0^{\frac{2}{3}}) \\ &= y_0^{\frac{2}{3}} (1 + y_0^{\frac{4}{3}}x_0^{\frac{2}{3}} + 2x_0^{\frac{2}{3}}y_0^{\frac{2}{3}} + x_0^{\frac{2}{3}}(1 + y_0^{\frac{4}{3}}x_0^{\frac{2}{3}} + 2y_0^{\frac{2}{3}}x_0^{\frac{2}{3}})) \\ &\sim ? = (x^{\frac{2}{3}} + y^{\frac{2}{3}})^3 \end{aligned}$$

例 5.7

设 $y=y(x)$ 是由参数方程 $\begin{cases} x=2t+t^2 \\ y=5t^2+4t \end{cases}$ 所确定.

在 $t=0$ 处, 函数 $y=y(x)$

$$\text{[分析]} \quad \frac{dy}{dx} = \frac{10t+4}{2+1} = \frac{10t}{3} \quad t \neq 0.$$

$$\frac{10t+4t-4t}{2-1} = \frac{10}{3}t, \quad t < 0.$$

$$\lim_{t \rightarrow 0^+} \frac{dy}{dx} > 0, \quad \lim_{t \rightarrow 0^-} \frac{dy}{dx} < 0.$$

导数不存在, χ

$$\begin{cases} x=3t & t \geq 0 \Rightarrow y=x \\ y=9t^2 & t < 0 \Rightarrow y=x^2 \end{cases}$$

$\Rightarrow x=0$ 时 可导性小.

例 5.8

求 $f(x) = |x|e^{-|x-1|}$ 的极值

$$f(x) = \begin{cases} xe^{-(x-1)}, & x \geq 1 \\ xe^{(x-1)}, & x \in (0, 1) \\ -xe^{(x-1)}, & x < 0 \end{cases}$$

$$x \geq 0 \quad x < 0$$

$$x \geq 1 \quad x < 1$$

$$f'(x) = \begin{cases} e^{1-x}(1-x), & x \geq 1 \\ e^{x-1}(x+1), & x \in (0, 1) \\ -e^{x-1}(x+1), & x < 0 \end{cases}$$

$$\lim_{x \rightarrow 1^+} f'(x) = 0, \quad \lim_{x \rightarrow 0^-} f'(x) = -e^{-1}$$

$$\lim_{x \rightarrow 1^-} f'(x) = 0, \quad \lim_{x \rightarrow 0^+} f'(x) = e^{-1}$$

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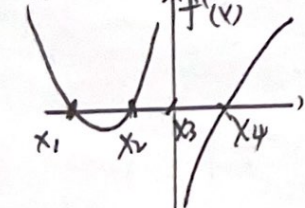
$$\lim_{x \rightarrow 0^+} f'(x) = e^{-1}$$

x	$(-\infty, -1)$	-1	$(-1, 0)$	0	$(0, 1)$	1	$(1, +\infty)$
f(x)	↗	e^{-2}	↘	0	↗	1	↘
f'(x)	+	0	-	0	+	0	-

$f(-1) = e^{-2}$ 是极大, $f(0) = 0$ 是极小, $f(1) = 1$ 是极大.

例 5.9

设 $f(x)$ 在 $(-\infty, +\infty)$ 内连续, (2) 个极小, (2) 个极大.



[分析] $f'(x) = 0$ 的点.

x_1 极大 x_2 极小 x_3 极大 x_4 极小

x_1 极大 x_2 极小 x_3 极大 x_4 极小

x_1 极大 x_2 极小 x_3 极大 x_4 极小

x_1 极大 x_2 极小 x_3 极大 x_4 极小

例 5.10. ~~xxxx~~

$$f(x) = \int_0^x \frac{(t+3)(t-1)}{e^t \sqrt{1+t^2}} dt, \text{ 求 } f(x) = ? \text{ 极大? 极小.}$$

$$f'(x) = \frac{(x+3)(x-1)}{e^x \sqrt{1+x^2}} \stackrel{?}{=} 0. \quad t_1 = -3 \quad t_2 = 1 \quad t_3 = -1$$

$\Rightarrow$ 2 极小 1 极大

例 5.11

求 $f(x) = \int_1^x (x^2 - t) e^{-t^2} dt$ 的单调区间与极值.

$$= \int_1^x x^2 e^{-t^2} dt - \int_1^x t e^{-t^2} dt$$

$$= x^2 \int_1^x e^{-t^2} dt - \int_1^x t e^{-t^2} dt$$

$$f'(x) = x^2 e^{-x^2} \cdot 2x + \int_1^x e^{-t^2} dt - 2x \cdot x^2 e^{-x^2}$$

$$= 2x \int_1^x e^{-t^2} dt \stackrel{?}{=} 0. \Rightarrow x=0, x=\pm 1$$

$$x \quad (-\infty, -1) \quad -1 \quad (-1, 0) \quad 0 \quad (0, 1) \quad 1 \quad (1, +\infty)$$

$$f'(x) \quad - \quad 0 \quad + \quad 0 \quad - \quad 0 \quad +$$

$$f(x) \quad \searrow \text{极大} \quad \nearrow \text{极小} \quad \searrow \text{极大} \quad \nearrow \text{极小} \quad \searrow \text{极大} \quad \nearrow$$

$$+ \int_0^1 t e^{-t^2} dt = \int_0^1 e^{-u} du = \frac{1}{2}(1 - e^{-1})$$

例 5.12.

$$x^2 + y^2 - 3x + 3y - 2 = 0. \text{ 求 } y(x) \text{ 极值}$$

$$2x + 2y \cdot y' - 3 + 3 \cdot y' = 0 \Rightarrow y' = \frac{1-x^2}{1+y^2} \stackrel{?}{=} 0 \Rightarrow x = \pm 1$$

图形

例 5.13.

设 $f(x)$: $xf''(x) + 3x[f'(x)]^2 = 1 - e^{-x}$, $f''(x)$

(1). 如果 $f(x)$ 在点 $x=c$ ($c \neq 0$) 处有极值, 证明它是极小值.

$$f'(c) = 0.$$

$$cf''(c) + 3c \cdot 0 = 1 - e^{-c}$$

$$\Rightarrow f''(c) = \frac{1 - e^{-c}}{c}$$

$c > 0$: $f''(c) > 0$
 $c < 0$: $f''(c) > 0 \Rightarrow$ 极小值

(2). 如果 $f(x)$ 在 $x=0$ 处有极值, 极大, 小?

$$\lim_{x \rightarrow 0} f''(x) = \lim_{x \rightarrow 0} \frac{f'(x) = 0}{1} = 1 > 0 \Rightarrow \text{极小值}$$

例 5.15.

$f(x)$ 二阶可导, $g(x) = f(0)(1-x) + f(1)x$.
 在 $[0, 1]$ 上

[分析]. $f(\lambda_1 x_1 + \lambda_2 x_2) \sim \lambda_1 f(x_1) + \lambda_2 f(x_2)$.

$$g(x) = f(0) \underbrace{(1-x)}_{\lambda_1} + f(1) \underbrace{x}_{\lambda_2} \sim f((1-x)0 + 1 \cdot x) = f(x).$$

当 $f''(x) \geq 0$, $f(x) \leq g(x)$.



例 5.14.

$y(x) = \begin{cases} e^{\frac{1}{x}}, & x < 0 \\ (3-x)\sqrt{x}, & x > 0 \end{cases}$ 凹凸区间与拐点.

$$y'(x) = \begin{cases} e^{\frac{1}{x}} \cdot \frac{1}{x^2}, & x < 0 \\ -\sqrt{x} + \frac{3-x}{2} x^{-\frac{1}{2}}, & x > 0 \end{cases}$$

$$y''(x) = \begin{cases} e^{\frac{1}{x}} \cdot \frac{1}{x^3} + e^{\frac{1}{x}} x^{-3} = e^{\frac{1}{x}} \cdot \frac{2}{x^4}, & x < 0 \\ -\frac{1}{2} x^{-\frac{1}{2}} + \frac{1}{4} (3-x) x^{-\frac{3}{2}} = -\frac{1}{4} (x^{-\frac{1}{2}} + x^{-\frac{3}{2}}), & x > 0 \end{cases}$$

$$\text{令 } y''(x) = 0 \quad x = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} y'(x) = 0. \Rightarrow f'(0) \text{ 不存在}$$

$$\lim_{x \rightarrow 0^-} y'(x) = +\infty.$$

$$y''(x) < 0 \quad \text{当 } x \in (-\infty, -\frac{1}{2})$$

$$x \in (-\frac{1}{2}, 0) \quad \text{凹}$$

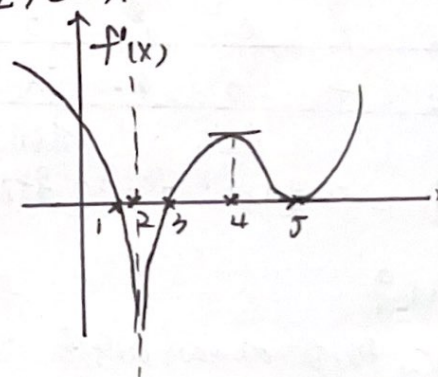
拐点: $(0, 0), (-\frac{1}{2}, e^{-2})$.

例 5.16.

$f(x)$ 在 $(-\infty, +\infty)$ 连续.

$$\begin{cases} f'(x) = 0 & x_1, x_3, x_5 \\ \text{不存} & x_2 \end{cases}$$

$$\begin{cases} f''(x) = 0 & x_4, x_5 \\ \text{不存} & x_2 \end{cases}$$



极值点 ($f'(x)$ 变号) $\Rightarrow x_1, x_3, x_5$ 2个

拐点 ($f''(x)$ 变号) $\Rightarrow x_2, x_4, x_5$ 3个

例 5.17

$$f(x) = e^{\frac{1}{x}} \sqrt{x^2 - 4x + 5} + x \left[\frac{1}{x} \right] \quad \text{渐近线方程?}$$

$$= e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} + x \left[\frac{1}{x} \right]$$

step 1 定义域: $x \neq 0$.

① $x > 1$. $0 < \frac{1}{x} < 1$. $\left[\frac{1}{x} \right] = 0$.

$$\lim_{x \rightarrow +\infty} \left[\frac{1}{x} \right] = 0 \Rightarrow \lim_{x \rightarrow +\infty} x \left[\frac{1}{x} \right] = 0.$$

② $x \leq -1$. $-1 \leq \frac{1}{x} < 0$. $\left[\frac{1}{x} \right] = -1$

$$\lim_{x \rightarrow -\infty} \left[\frac{1}{x} \right] = -1 \Rightarrow \lim_{x \rightarrow -\infty} x \left[\frac{1}{x} \right] = +\infty.$$

③ $x \in (0, 1)$. $\frac{1}{x} - 1 \leq \left[\frac{1}{x} \right] \leq \frac{1}{x} \Rightarrow 1 - x < x \left[\frac{1}{x} \right] \leq 1$

$$\lim_{x \rightarrow 0} x \left[\frac{1}{x} \right] = 1 \Rightarrow \lim_{x \rightarrow 0} x \left[\frac{1}{x} \right] = 1. \quad (\text{夹逼})$$

step 2.

(1) 铅垂: $\lim_{x \rightarrow 0^+} \left\{ e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} + x \left[\frac{1}{x} \right] \right\} = +\infty$. $x=0$ 是一条铅垂

(2) 水平: $\lim_{x \rightarrow +\infty} \left\{ e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} + x \left[\frac{1}{x} \right] \right\} = +\infty$ 无水平

$$\lim_{x \rightarrow -\infty} \left\{ e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} + x \left[\frac{1}{x} \right] \right\} = +\infty$$

(3) 斜:

$$\lim_{x \rightarrow +\infty} \frac{e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} + x \left[\frac{1}{x} \right]}{x} = \lim_{x \rightarrow +\infty} \left[\frac{e^{\frac{1}{x}} \sqrt{(x-1)(x+1)}}{x} + \left[\frac{1}{x} \right] \right] = 1$$

$$\lim_{x \rightarrow +\infty} f(x) - x = \lim_{x \rightarrow +\infty} \left[e^{\frac{1}{x}} \sqrt{(x-1)(x+1)} - x \right] = -1$$

$$y = x - 1$$

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} \sqrt{1 - \frac{4}{x} + \frac{5}{x^2}} + \lim_{x \rightarrow -\infty} \left[\frac{1}{x} \right] = \lim_{x \rightarrow -\infty} \frac{e^{\frac{1}{x}} \sqrt{x^2 - 4x + 5}}{-x} = -2$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} f(x) + 2x &= \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} \sqrt{x^2 - 4x + 5} + x \left[\frac{1}{x} \right] + 2x \\ &= \lim_{x \rightarrow -\infty} \left[e^{\frac{1}{x}} \sqrt{x^2 - 4x + 5} + x \right] + x \left[\frac{1}{x} + 1 \right] \\ &= \lim_{x \rightarrow -\infty} \left[-x \left(e^{\frac{1}{x}} \sqrt{1 - \frac{4}{x} + \frac{5}{x^2}} - 1 \right) + e^{\frac{1}{x}} - e^{\frac{1}{x}} \right] \\ &= \lim_{x \rightarrow -\infty} \left(-x \left(e^{\frac{1}{x}} \left(-\frac{1}{2} \right) \left(\frac{5}{x^2} - \frac{4}{x} \right) + e^{\frac{1}{x}} - 1 \right) \right) \\ &= -(-2+1) = 1 \end{aligned}$$

例 5.18.

求曲线 $x^3 + y^3 = 3xy$ 的斜渐近线方程

1) 设 $y = ax + b$.

$$a = \lim_{x \rightarrow \infty} \frac{y}{x} \quad \text{令 } \frac{y}{x} = u \quad \text{则 } y = ux$$

$$x^3 + u^3 x^3 = 3ux^2 \Rightarrow 1 + u^3 = \frac{3u}{x}$$

$$\Rightarrow u^3 = \frac{3u}{x} - 1 \Rightarrow \lim_{x \rightarrow \infty} \frac{3u}{x} - 1 = -1 \Rightarrow a = -1$$

2) $b = \lim_{x \rightarrow \infty} y - ax = \lim_{x \rightarrow \infty} y + x$ 令 $y + x = t$ 则 $y = t - x$

$$\Rightarrow x^3 + (t-x)^3 = 3x(t-x) \Rightarrow t^3 + 3xt - 3tx = 3xy - 3x^2$$

$$\Rightarrow \frac{t^3}{x^2} + \frac{3t}{x} - \frac{3t}{x} = \frac{3xt}{x} - 3 \Rightarrow t = \sqrt[3]{\frac{3xt + 3t^2 - t^3}{x}} \rightarrow -1$$

$$3) \Rightarrow y = -x - 1$$

例 5.20

设 $f(x) = \int_x^{x+\frac{\pi}{2}} |\sin t| dt$

1. 证明 $f(x)$ 是以 π 为周期的周期函数.

2. 求 $f(x)$ 的值域.

1. $f(x+\pi) = \int_{x+\pi}^{x+\pi+\frac{\pi}{2}} |\sin t| dt = \int_{x+\pi}^{x+\pi+\frac{\pi}{2}} |\sin(t+\pi)| dt$

$\begin{matrix} x+t+\pi=u \\ t=u-\pi \end{matrix} \int_x^{x+\frac{\pi}{2}} |\sin u| du = \int_x^{x+\frac{\pi}{2}} |\sin t| dt$

2. 由 1) 只需研究 $[0, \pi]$ 即可

$f(x) = 0 \quad f(x) = |\sin(x+\frac{\pi}{2})| - |\sin x|$
驻点: $= |\cos x| - |\sin x| = 0 \Rightarrow x_1 = \frac{\pi}{4} \quad x_2 = \frac{3}{4}\pi$

端点: $x_3 = 0 \quad x_4 = \pi$

$f(\frac{\pi}{4}) = \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} |\sin t| dt = 2 \quad f(\frac{3}{4}\pi) = \int_{\frac{3}{4}\pi}^{\pi} |\sin t| dt = 2 - \sqrt{2}$

$f(0) = \int_0^{\frac{\pi}{2}} |\sin t| dt = 1 \quad f(\pi) = \int_{\pi}^{\frac{3}{2}\pi} |\sin t| dt = -1$



$\therefore [2-\sqrt{2}, 2]$

例 5.21

$2y^3 - 2y^2 + 2xy - x^2 = 1$ 在 $(1, 1)$ 处求切线?

[solution] $6y^2 \cdot y' - 4y \cdot y' + 2y + 2x y' - 2x = 0$
 $\Rightarrow y' = \frac{2x-2y}{2x+6y^2-4y} = \frac{x-y}{x+3y^2-2y} \quad x=1, y=1 \Rightarrow y'=0$

$y'' = \frac{(1-y')(x+3y^2-2y) - (x-y)(1+6y \cdot y' - 2y')}{(x+3y^2-2y)^2}$

$x=1, y=1, y'=0 \Rightarrow \frac{(1-0)(1+3-2) - 0}{(1+3-2)^2} = \frac{1}{2}$

$k = \frac{1}{k} = \frac{1}{\frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}}} = 2$

例 5.22

求心形线 $r = a(1 + \cos \theta)$ ($a > 0$) 在 $M(\frac{\pi}{2}, a)$ 处切线.

$\frac{dy}{dx} = \frac{d[a \sin \theta (1 + \cos \theta)] / d\theta}{d[a \cos \theta (1 + \cos \theta)] / d\theta} = \frac{(\cos \theta + \cos 2\theta)}{(-\sin \theta + \sin 2\theta)} \bigg|_{\frac{\pi}{2}} = 1$

$\frac{d(\frac{dy}{dx})}{dx} = \frac{(\frac{\cos \theta + \cos 2\theta}{-\sin \theta + \sin 2\theta})' \bigg|_{\frac{\pi}{2}}}{(-\cos \theta + \cos 2\theta)' \bigg|_{\frac{\pi}{2}}} = -\frac{3}{a}$

$k = \frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}} = \frac{3}{2\sqrt{2}a}$

例 5.22

求曲线 $y = \ln x$ 上曲率最大点, 并在该点附近用 $y = ax^2 + bx + c$ 近似代替 $y = \ln x$, 求 a, b, c .

$$① \quad y' = \frac{1}{x} \quad y'' = -\frac{1}{x^2}$$

$$k = \frac{|1 - \frac{1}{x^2}|}{(1 + \frac{1}{x^2})^{\frac{3}{2}}} = \frac{\frac{1}{x^2}}{(1 + \frac{1}{x^2})^{\frac{3}{2}}} = \frac{x}{(x^2 + 1)^{\frac{3}{2}}} = \left(\frac{1}{x^{\frac{1}{2}} + x^{\frac{3}{2}}} \right)^{\frac{3}{2}}$$

$$k'(x) = -\frac{3}{2} \left(\frac{1}{x^{\frac{1}{2}} + x^{\frac{3}{2}}} \right)^{\frac{3}{2}} \cdot \frac{-\frac{1}{2}x^{-\frac{1}{2}} + \frac{3}{2}x^{\frac{1}{2}}}{(x^{\frac{1}{2}} + x^{\frac{3}{2}})^2} = -\frac{3}{4} \frac{-x^{\frac{1}{2}} + 3x^{\frac{3}{2}}}{(x^{\frac{1}{2}} + x^{\frac{3}{2}})^{\frac{5}{2}}}$$

$$\text{令 } k'(x) = 0 \Rightarrow \frac{1}{x^{\frac{1}{2}}} = \frac{3}{2}x^{\frac{1}{2}} \Rightarrow x = \frac{\sqrt{2}}{2}$$

$$\text{当 } 0 < x < \frac{\sqrt{2}}{2}, k' < 0; \text{当 } x > \frac{\sqrt{2}}{2}, k' > 0.$$

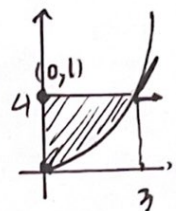
$$\Rightarrow k_{\max} = k\left(\frac{\sqrt{2}}{2}\right)$$

$$② \quad y_1 = y_2 \quad y'_1 = y'_2 \quad |y''_1| = |y''_2| \quad (\Leftarrow k_1 = k_2)$$

$$\begin{cases} \ln x = ax^2 + bx + c \\ \frac{1}{x} = 2ax + b \\ -\frac{1}{x^2} = 2a \end{cases} \quad \left| \begin{array}{l} x = \frac{\sqrt{2}}{2} \\ \Rightarrow \end{array} \right. \begin{cases} a = -1 \\ b = 2\sqrt{2} \\ c = -\frac{1}{2}(\ln 2 + 3) \end{cases}$$

例 5.24

已知 $L: y = \frac{4}{9}x^2 (x \geq 0)$. $O(0,0)$ $A(0,1)$
 P 点... S 是 OA 与 AP 及 L 所围面积,
 P 至 $(3,4)$ $dx=4$. 求 $\frac{dS}{dt}$.



$$P(m, \frac{4}{9}m^2) \quad \frac{dS}{dt} = \frac{dS}{dm} \cdot \frac{dm}{dt}, \quad \frac{4}{9}m^2 - 1$$

$$y - 1 = k(x - 0) \quad k = \frac{\frac{4}{9}m^2 - 1}{m - 0}$$

$$y = (\frac{4}{9}m - \frac{1}{m})x + 1$$

$$S = \int_0^m \left[\left(\frac{4}{9}m - \frac{1}{m} \right)x + 1 \right] dx$$

$$= \frac{2}{27}m^3 + \frac{m}{2}$$

$$\frac{dS}{dt} = \left(\frac{2}{9}m^2 + \frac{1}{2} \right) \frac{dm}{dt} = \frac{8}{9}m^2 + 2 \quad m=3 \text{ 时 } \frac{dS}{dt} = 10.$$

重点 ★ ★ ★ ★ ★ ★
 5.2 5.4 5.5 5.8 5.10 5.14

第五讲 习题与1000题总结.

5.2 曲线L极坐标方程, $r=\theta$, L在 $(1, \theta)=(\frac{\pi}{2}, \frac{\pi}{2})$

处取切线的直角坐标方程是_____

$$\begin{cases} x=r\cos\theta=\theta\cos\theta \\ y=r\sin\theta=\theta\sin\theta \end{cases} \quad \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin\theta + \theta\cos\theta}{\cos\theta - \theta\sin\theta} = k$$

$$k(\frac{\pi}{2}) = \frac{1+0}{0-\frac{\pi}{2}} = -\frac{2}{\pi}, \quad x(\frac{\pi}{2})=0 \quad y(\frac{\pi}{2})=\frac{\pi}{2}$$

$$\text{切线: } y - \frac{\pi}{2} = -\frac{2}{\pi}x.$$

5.4 设 $y=\tan^n x$ 在 $x=\frac{\pi}{4}$ 处的切线X轴截距为 x_1 , 求 $\lim_{n \rightarrow \infty} y(x_n)$.

$$1. y' = n \tan^{n-1} x \cdot \frac{1}{\cos^2 x} \Big|_{x=\frac{\pi}{4}} = 2n.$$

$$\text{切线: } y-1 = 2(x-\frac{\pi}{4}) \quad y=0, \quad x_1 = \frac{\pi}{4} - \frac{1}{2n}$$

$$\begin{aligned} 2. \lim_{n \rightarrow \infty} y(x_n) &= \lim_{n \rightarrow \infty} \tan^n(\frac{\pi}{4} - \frac{1}{2n}) = \lim_{n \rightarrow \infty} \left[\frac{1 - \tan \frac{1}{2n}}{1 + \tan \frac{1}{2n}} \right]^n \\ &= \lim_{n \rightarrow \infty} \frac{[(1 - \tan \frac{1}{2n})^{-\frac{1}{\tan \frac{1}{2n}}}]^{n \tan \frac{1}{2n}}}{[(1 + \tan \frac{1}{2n})^{-\frac{1}{\tan \frac{1}{2n}}}]^{n \tan \frac{1}{2n}}} \rightarrow -\frac{1}{2} \\ &= \frac{e^{-\frac{1}{2}}}{e^{\frac{1}{2}}} = e^{-1} \end{aligned}$$

5.5 设 $f(x)$ 有连续的一阶导数, 且 $f(0)=0, f'(0)=1$. 求 $\lim_{x \rightarrow 0} \frac{x f(x)}{u f(x)}$
其中 u 是曲线 $y=f(x)$ 在 $(x, f(x))$ 处的切线在X轴上的截距,

$$f(0)=0 \quad f'(0)=1.$$

$$y - y(x_0) = f'(x_0) \cdot (x - x_0)$$

$$x \rightarrow 0, \quad y - 0 = 1(u - 0) \Rightarrow y(0) = u(0).$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x f(x)}{u f(x)} &= \lim_{x \rightarrow 0} \frac{x}{f(x)} \cdot \lim_{u \rightarrow 0} \frac{f(u)}{u} \\ &= \lim_{x \rightarrow 0} \frac{x-0}{f(x)-f(0)} \cdot \lim_{u \rightarrow 0} \frac{f(u)-f(0)}{f(u)-0} \\ &= \frac{f'(x)}{f'(x)} = 1. \end{aligned}$$

5.8 设 $f(x) = a|\cos x| + b|\sin x|$

在 $x = -\frac{\pi}{3}$ 处取得极小, 且 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [f(x)]^2 dx = 2$, 求 a, b .

$$x \in (-\frac{\pi}{2}, 0): f(x) = a\cos x - b\sin x$$

$$f'(x) = -a\sin x - b\cos x$$

$$f'(-\frac{\pi}{3}) = \frac{+a\sqrt{3}}{2} - \frac{b}{2} = 0 \Rightarrow a\sqrt{3} - b = 0$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [f(x)]^2 dx = 2 \int_0^{\frac{\pi}{2}} (a\cos x + b\sin x)^2 dx = 2 \quad b = \sqrt{3}a$$

$$\int_0^{\frac{\pi}{2}} a^2 \cos^2 x + b^2 \sin^2 x + 2ab \cos x \sin x = 1$$

$$\int_0^{\frac{\pi}{2}} \frac{a^2+b^2}{2} + \frac{a^2}{2} \cos 2x - \frac{b^2}{2} \sin 2x + ab \sin 2x$$

$$= \frac{\pi}{2} \frac{a^2+b^2}{2} + \frac{a^2}{4} \sin 2x \Big|_0^{\frac{\pi}{2}} + \frac{b^2}{4} \cos 2x \Big|_0^{\frac{\pi}{2}} + \frac{ab}{2} \cos 2x \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{a^2-2ab+b^2\pi+a^2\pi}{4} \Rightarrow a = -\sqrt{\frac{1}{b+\pi}}, \quad b = -\sqrt{\frac{3}{b+\pi}}$$

5.12 ~~★★~~

$f(x) = nx(1-x)^n$ (n 为正整数), 求:

(1) $f(x)$ 在 $[0, 1]$ 上最大值 $M(n)$.

(2) $\lim_{n \rightarrow \infty} M(n)$.

$$f'(x) = n(1-x)^n - n^2x(1-x)^{n-1} \stackrel{?}{=} 0.$$

$$\Rightarrow x = \frac{1}{n+1}. \quad x \in (0, \frac{1}{n+1}), f'(x) > 0, f(x) \uparrow$$

$$x \in (\frac{1}{n+1}, 1), f'(x) < 0, f(x) \downarrow.$$

$$f(\frac{1}{n+1}) = \frac{n}{n+1} (\frac{n}{n+1})^n = (\frac{n}{n+1})^{n+1}$$

$$(2) \lim_{n \rightarrow \infty} M(n) = \lim_{n \rightarrow \infty} (1 - \frac{1}{n+1})^{n+1} = \lim_{x \rightarrow \infty} (1 + \frac{-1}{x})^x = e^{-1}$$

1000.5.4

周期函数 $f(x)$ 在 $(-\infty, +\infty)$ 内可导, 周期为 4.

$$\lim_{x \rightarrow 0} \frac{f(1) - f(1-x)}{2x} = -1. \text{ 则 } f'(1) = ?$$

$$\lim_{x \rightarrow 0} \frac{f(1) - f(1-x)}{x} \cdot \frac{1}{2} = -1 \Rightarrow f'(1) = f'(1) = -2$$

1000.5.7

$f(x)$ 有连续导数, $f(x) + \int_0^x f(t) dt = \frac{3}{2}x^2 + \frac{2}{3}$

则 $f(x)$ 必存在极大/小值()?

(solution).

$$f(x) + \int_0^x f(t) dt = \frac{3}{2}x^2 + \frac{2}{3}$$

常微分方程

$$f'(x) + 3f(x) = 3x$$

$$\begin{cases} y' + P(x)y = Q(x) \\ \Rightarrow y = C e^{-\int P(x)dx} + e^{-\int P(x)dx} \int Q(x) e^{\int P(x)dx} dx \end{cases}$$

$$\Rightarrow y = C e^{-3x} + e^{-3x} \int 3x e^{3x} dx$$

$$y = (e^{-3x} + x - \frac{1}{3}) e^{-3x} \Rightarrow C = 1$$

$$\Rightarrow y = e^{-3x} + x - \frac{1}{3}, \quad y' = -3e^{-3x} + 1 \stackrel{?}{=} 0.$$

$$f'(x) = 3f(x) = 3 > 0.$$

$$f(x) = \frac{1}{3} \ln 3.$$

$\therefore$ 极大值, $f(x) = \frac{1}{3} \ln 3.$

1000.5.8

拐点个数?

$$f(x) = \begin{cases} e^x, & x < 0. \\ (4-x)\sqrt{x}, & x > 0. \end{cases}$$

(solution).

$$f'(x) = \begin{cases} e^x, & x < 0. \\ \frac{3(4-x)}{2\sqrt{x}}, & x > 0. \end{cases} \quad f''(x) = \begin{cases} e^x(1+2x), & x < 0. \\ \frac{-3(4-x)}{4x\sqrt{x}}, & x > 0. \end{cases}$$

$$\text{令 } f''(x) = 0 \Rightarrow x = -\frac{1}{2}.$$

$$x \in (-\frac{1}{2}, 0) \text{ 时, } f''(x) > 0. \quad x \in (0, +\infty) \text{ 时 } f''(x) < 0.$$

$x=0$ 也是拐点. $\Rightarrow 2$

1000. S.16 ???!

$$f(x) = \int_0^1 (x-t) dt + \int_0^x \sqrt{x^2-t^2} dt \quad (0 < x < +\infty).$$

[solution].

$$f(x) = \begin{cases} \frac{\pi+4}{4} x^2 - x + \frac{1}{2}, & 0 < x < 1 \\ \frac{\pi}{4} x^2 + x - \frac{1}{2}, & x > 1. \end{cases} \quad ?$$

$$f'(x) = \begin{cases} \frac{\pi+4}{2} x - 1 \\ \frac{\pi}{2} x + 1 \end{cases}$$

$$\text{令 } f'(x) = 0, \Rightarrow x_0 = \frac{2}{\pi+4}.$$

$$f''(x_0) = \frac{\pi+4}{2} > 0, \quad f(x_0) = \frac{1}{2} - \frac{1}{\pi+4} \text{ 极小值} \rightarrow \text{最小值}$$

$$\lim_{x \rightarrow +\infty} f(x) = +\infty, \Rightarrow \text{无最大值}$$

1000. S.18.

曲线 $y = \frac{x^2 \arctan x}{x-1}$ 渐近线条数

$$\lim_{x \rightarrow \infty} \frac{x^2 \arctan x}{x-1} = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{x^2 \arctan x}{x(x-1)} = \frac{\pi}{2} \rightarrow \text{水平一条}$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 \arctan x}{x(x-1)} = -\frac{\pi}{2} \rightarrow \text{水平一条}$$

$$\lim_{x \rightarrow 1} \frac{x^2 \arctan x}{x-1} = \infty \rightarrow \text{铅垂一条}$$

3条

1000. S.19

$$f(x) = e^x + \sqrt{x^2+x+1} + \sqrt{x^2-x+1} \quad \text{渐近线条数.}$$

[solution]

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{e^x}{x} + \lim_{x \rightarrow +\infty} \frac{x(\sqrt{1+\frac{1}{x}+\frac{1}{x^2}} + \sqrt{1-\frac{1}{x}+\frac{1}{x^2}})}{x} = 2.$$

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) - 2x &= \lim_{x \rightarrow +\infty} e^x + \lim_{x \rightarrow +\infty} x(\sqrt{1+\frac{1}{x}+\frac{1}{x^2}} + \sqrt{1-\frac{1}{x}+\frac{1}{x^2}} - 2) \\ &= \lim_{x \rightarrow +\infty} 1 + \frac{1}{2} \frac{(\frac{1}{x} + \frac{1}{x^2}) - (\frac{1}{x} - \frac{1}{x^2})}{\frac{1}{x}} = 1 \end{aligned}$$

斜渐近线 $y = 2x + 1$

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \lim_{x \rightarrow -\infty} \frac{(-x)(\sqrt{1+\frac{1}{x}+\frac{1}{x^2}} + \sqrt{1-\frac{1}{x}+\frac{1}{x^2}})}{x} = -2$$

$$\lim_{x \rightarrow -\infty} f(x) + 2x = 1.$$

斜渐近线 $y = -2x + 1$

铅垂渐近线 $x = 0$

注意根号提出 $(-x)$

1000. S.22

$$y = \sqrt{4x^2 - 3x + 7} - 2x \Rightarrow \text{渐近线条数.}$$

[solution].

$$y = \sqrt{4x^2 - 3x + 7} - 2x \Rightarrow$$

$$\lim_{x \rightarrow +\infty} 2x(\sqrt{1 - \frac{3}{4x} + \frac{7}{4x^2}} - 1) = \lim_{x \rightarrow +\infty} (2x \cdot \frac{1}{2})(\frac{7}{4x^2} - \frac{3}{4x})$$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{f(x) - 2x}{x} &= \lim_{x \rightarrow +\infty} \frac{\sqrt{4x^2 - 3x + 7} - 2x}{x} = \lim_{x \rightarrow +\infty} \frac{-\sqrt{1 - \frac{3}{4x} + \frac{7}{4x^2}}}{1} = -\frac{3}{4} \\ \lim_{x \rightarrow +\infty} \sqrt{4x^2 - 3x + 7} - 2x + 4x &= \frac{3}{4} \Rightarrow \text{斜渐近线 } y = -4x + \frac{3}{4} \end{aligned}$$

1000.5t.5

$$f(x) = \max_{0 \leq y \leq 1} \frac{|x-y|}{x+y+1}, \quad 0 \leq x \leq 1, \text{ 则 } f(x) \text{ 在 } [0, 1] \text{ 上}$$

的 min, max = ?

① $0 < y \leq x$ 时

$$f(x) = \max_{0 \leq y \leq 1} \left(\frac{x-y}{x+y+1} \right) = \max_{0 \leq y \leq 1} \left(\frac{2x+1}{x+y+1} - 1 \right)$$

$$y=0 \Rightarrow \frac{2x+1}{x+1} - 1 = \frac{x}{x+1}$$

② $x < y \leq 1$ 时.

$$f(x) = \max_{0 \leq y \leq 1} \frac{y-x}{x+y+1} = \max_{0 \leq y \leq 1} \frac{2y+1}{x+y+1} - 1 = \frac{y}{x}$$

$$\frac{1}{x+2} - 1 = \frac{1-x}{x+2}$$

$$\text{③ } 0 < y \leq x \text{ 时 } \max \left\{ \frac{x}{x+1} \right\} = \frac{1}{2} \min \left\{ \frac{x}{x+1} \right\} = 0.$$

$$x < y \leq 1 \text{ 时 } \max \left\{ \frac{1}{x+2} - 1 \right\} = 0.5 \min \left\{ \frac{1}{x+2} - 1 \right\} = 0.$$

$$f(x) = \max_{0 \leq y \leq 1} \left\{ \frac{1-x}{x+2}, \frac{x}{x+1} \right\}$$

$$\text{注意到: } \frac{x}{x+1} \leq \frac{1-x}{x+2} \Leftrightarrow (x+\frac{1}{2})^2 \leq \frac{3}{4} \Leftrightarrow 0 < x \leq \frac{\sqrt{3}-1}{2}$$

啥!??

1000.5t.6.

设 $f(x)$ 在 $x=0$ 的某邻域内连续在 $x=0$ 处存在

一阶导数 $f'(0)$. 又设

$$\lim_{x \rightarrow 0} \frac{\int_0^x t f(x-t) dt}{x^4} = a \quad (\text{常数 } a > 0).$$

求 $x=0$.

[solution]. $x-t=u$

$$\lim_{x \rightarrow 0} \frac{\int_0^x t f(x-t) dt}{x^4} = \lim_{x \rightarrow 0} \frac{\int_0^x (x-u) f(u) du}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{x \int_0^x f(u) du}{x^4} + \lim_{x \rightarrow 0} \frac{\int_0^x u f(u) du}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{x f(x) - \int_0^x f(u) du - x f(x)}{4x^3}$$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{12x^2} = a.$$

$$\frac{f'(x)}{24x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{f''(x)}{24} = a > 0.$$

$\therefore x=0$ 是极小值点

1000.5t.13

求 $y = \sqrt{4x^2+x} \ln(2+\frac{1}{x})$ 的全部渐近线.

$$\text{① } \lim_{x \rightarrow +\infty} y = +\infty$$

$$\lim_{x \rightarrow +\infty} \frac{y}{x} = \lim_{x \rightarrow +\infty} \sqrt{4+\frac{1}{x}} \ln(2+\frac{1}{x}) = 2 \ln 2$$

$$\lim_{x \rightarrow +\infty} y - 2 \ln 2 x = \lim_{x \rightarrow +\infty} 2x \left(\sqrt{1+\frac{1}{4x}} \ln(2+\frac{1}{x}) - \ln 2 \right)$$

$$= \lim_{x \rightarrow +\infty} 2x \ln 2 \left(\sqrt{1 + \frac{1}{4x}} - 1 \right)$$

$$= \lim_{x \rightarrow +\infty} 2 \ln 2 \cdot \frac{1}{4x} = \frac{\ln 2}{4}$$

$$t = \frac{1}{x} \lim_{t \rightarrow 0^+} \frac{\sqrt{4+t} \ln(1+t) - 2 \ln 2}{t} \approx \lim_{x \rightarrow 0^+} \frac{\ln(1+t)}{2\sqrt{4+t}} + \frac{\sqrt{4+t}}{2+t}$$

$$= \frac{\ln 2}{4} + 1 \rightarrow \text{斜率 } y = 2 \ln 2 x + \frac{\ln 2}{4} + 1$$

$$\textcircled{2} \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = -2 \ln 2$$

$$\lim_{x \rightarrow -\infty} f(x) + 2 \ln 2 \rightarrow = \frac{\ln 2}{4} - 1$$

$$\rightarrow \text{斜率 } y = -2 \ln 2 x - \left(\frac{\ln 2}{4} + 1 \right)$$

$$\textcircled{3} x=0. \text{ 斜率}$$

1000.5.27

$$\begin{cases} x = 3t^2 + 2t + 3 \\ y = e^y \sin t + 1 \end{cases} \quad y = y(x) \text{ 在 } t=0 \text{ 对应的点处} \\ \text{求 } k = ?$$

$$\frac{dx}{dt} = 6t + 2 \quad \frac{dx}{dt} \Big|_{t=0} = 2$$

$$\frac{dy}{dt} = e^y y' \sin t + e^y \cos t \Rightarrow \frac{dy}{dt} = \frac{e^y \cos t}{1 - e^y \sin t}$$

$$\frac{dy}{dt} \Big|_{t=0} = e$$

$$u = \frac{dy}{dx} = \frac{e^y y' \sin t + e^y \cos t}{(1 - e^y \sin t)(6t + 2)} \Rightarrow \frac{dy}{dx} \Big|_{t=0} = \frac{e}{2}$$

$$\frac{d \frac{dy}{dx}}{dx} = \frac{dy/dt}{dx/dt}$$

$$(e^y \sin t)' = e^y y' \sin t + e^y \cos t$$

$$\frac{du}{dt} = \frac{[e^y y' \cos t - e^y \sin t](1 - e^y \sin t)(6t + 2) - e^y \cos t [(6t + 2)e^y (-y' \sin t - \cos t) + 6(1 - e^y \sin t)]}{[(1 - e^y \sin t)(6t + 2)]^2}$$

$$\frac{du}{dt} \Big|_{t=0} = \frac{\frac{e^2}{2} \cdot 2 - e[-2e + 6]}{4} = \dots \frac{e^3 - 3e}{2^2}$$

$$\frac{dy'}{dx} = \frac{e^3 - 3e}{4}$$

$$k = \frac{(y'')}{(1 + y'^2)^{3/2}} = \frac{2e(2e - 3)}{(4 + e^2)^{3/2}}$$

1000.5.37

$f(x)$ 可导, 且满足 $xf(x) = f(-x) + 1$, $f(0) = 0$.

1) $f'(x)$?

$$\begin{cases} xf'(x) = f'(-x) + 1 \\ -xf'(-x) = f'(x) + 1 \end{cases} \Rightarrow f'(x) = \frac{x-1}{1+x^2}$$

2) 求函数 $f(x)$ 的极值.

$$f(x) \quad (-\infty, 1) \searrow \quad (1, +\infty) \nearrow$$

$$f(x) = \int_0^1 f'(t) dt = \int_0^1 \frac{x-1}{1+t^2} dt$$

$$= \int_0^1 x d \arctan t - \int_0^1 1 d \arctan t$$

$$= (x \arctan t \Big|_0^1 - \int_0^1 \arctan t dt) - (\arctan t \Big|_0^1 - 0)$$

$$= - \int_0^1 \arctan t dt = \left(\frac{1}{2} \ln |x^2 + 1| - x \arctan x \right) \Big|_0^1 = \frac{\ln 2}{2} - \frac{\pi}{4}$$

1000.5t.16.

$$\text{设 } f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{1}{n} (1 + \cos \frac{x}{n} + \cos \frac{2x}{n} + \dots + \cos \frac{(n-1)x}{n}) & x > 0 \\ 1 & x = 0 \\ f(-x) & x < 0. \end{cases}$$

1). 求 $f'(0)$ 2). 求 $f(x)$ 在 $[-\pi, \pi]$ 最大值.

1) $x > 0$ 时

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} (1 + \cos \frac{x}{n} + \dots + \cos \frac{(n-1)x}{n}) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \cos \frac{ix}{n} \\ &= \lim_{n \rightarrow \infty} \frac{1}{x} \left(\sum_{i=0}^{n-1} \cos \frac{ix}{n} \right) \cdot \frac{x}{n} = \lim_{n \rightarrow \infty} \frac{1}{x} \int_0^x \cos t dt \\ &= \lim_{n \rightarrow \infty} \frac{\sin x}{x} \end{aligned}$$

$x < 0$ 时

$$\begin{aligned} f(x) &= \frac{\sin x}{x} \\ \lim_{x \rightarrow 0^+} \frac{\frac{\sin x}{x} - 1}{x - 0} &= \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x^2} = 0. \\ \lim_{x \rightarrow 0^-} \frac{\frac{\sin x}{x} - 1}{x - 0} &= \lim_{x \rightarrow 0^-} \frac{\sin x - x}{x^2} = 0. \end{aligned} \Rightarrow f'(0) = 0.$$

$$\begin{aligned} \text{2). } f'(x) &= \frac{x \cos x - \sin x}{x^2} \stackrel{?}{=} 0 \\ \Rightarrow x_0 &= \frac{\pi}{2}. \end{aligned}$$

$f(x)$ 是偶函数. $(0, \pi) \cup (-\pi, 0) \nearrow f(0) \text{ max} = 1$

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设曲线弧 $y = \sin x$ ($0 < x < \pi$)

1). 求曲线弧最小曲率半径

$$\begin{aligned} y &= \sin x \quad y' = \cos x \quad R = \frac{(1+y'^2)^{\frac{3}{2}}}{|y''|} = \frac{(1+\cos^2 x)^{\frac{3}{2}}}{|\sin x|} \\ R'(x) &= \frac{-\frac{3}{2}(1+\cos^2 x)^{\frac{1}{2}}(2\cos x \sin x)|\sin x| - (1+\cos^2 x)^{\frac{3}{2}} \cos x}{(\sin x)^2} \\ &= \frac{-\frac{3}{2}(1+\cos^2 x)^{\frac{1}{2}} \sin 2x \cdot |\sin x| - \cos x (1+\cos^2 x)^{\frac{3}{2}}}{(\sin x)^2} \leq 0. \end{aligned}$$

$$\begin{aligned} \cos x &= t \quad A = 1 - 16 < 0. \\ &= \frac{(1+\cos^2 x)^{-\frac{1}{2}} \cos x (-4-t^2+t)}{(\sin x)^2} < 0. \quad t = \cos x \\ \text{则 } x &= \frac{\pi}{2} \end{aligned}$$

$$R(\frac{\pi}{2}) = 1.$$

2). 求曲线弧在曲线单位圆上一点处相切具有相同曲率和凹性的抛物线.

点. $(\frac{\pi}{2}, 1)$. $k=R=1$

$$y = ax^2 + bx + c. \quad y' = 2ax + b \quad y'' = 2a.$$

$$\begin{aligned} 1 &= \frac{|2a|}{[(a\pi + b)^2 + 1]} \quad \text{相同 } y, y', y'' \\ 1 &= \frac{\pi^2}{4} a + \frac{\pi}{2} b + c \quad \begin{cases} A(\frac{\pi}{2})^2 + B(\frac{\pi}{2}) + C = \sin \frac{\pi}{2} \\ 2A \frac{\pi}{2} + B = \cos \frac{\pi}{2} \\ 2A = -\sin \frac{\pi}{2} \end{cases} \end{aligned}$$

$$\Rightarrow y = -\frac{1}{2}x^2 + \frac{\pi}{2}x + 1 - \frac{\pi^2}{8}$$